

Ferrara® HM Ring

Remodelamento corneano progressivo até 9 months — evidence of continuous flattening beyond the conventional stabilization period.

N = 136 olhos

Brazilian Cohort

Follow-up 24m



Clinical Data — Temporal Progression

5.05 D

ΔKMAX (9M)

4.94 D

ΔKM (9M)

+42.8 μm

ΔPACHYMETRY

+0.79

ΔQ (9M)

Timeline — Progressive ΔKmax

Pré

Baseline

—

1m

30 days

+1.3 D

3m

3 months

+0.35 D

9m

9 months

+5.05 D

12m+

12+ months

+7.14 D

⚡ 93% of Kmax reduction occurs AFTER the 3rd month — active progressive stromal remodeling

Summary — Freedom Cohort (all HM)

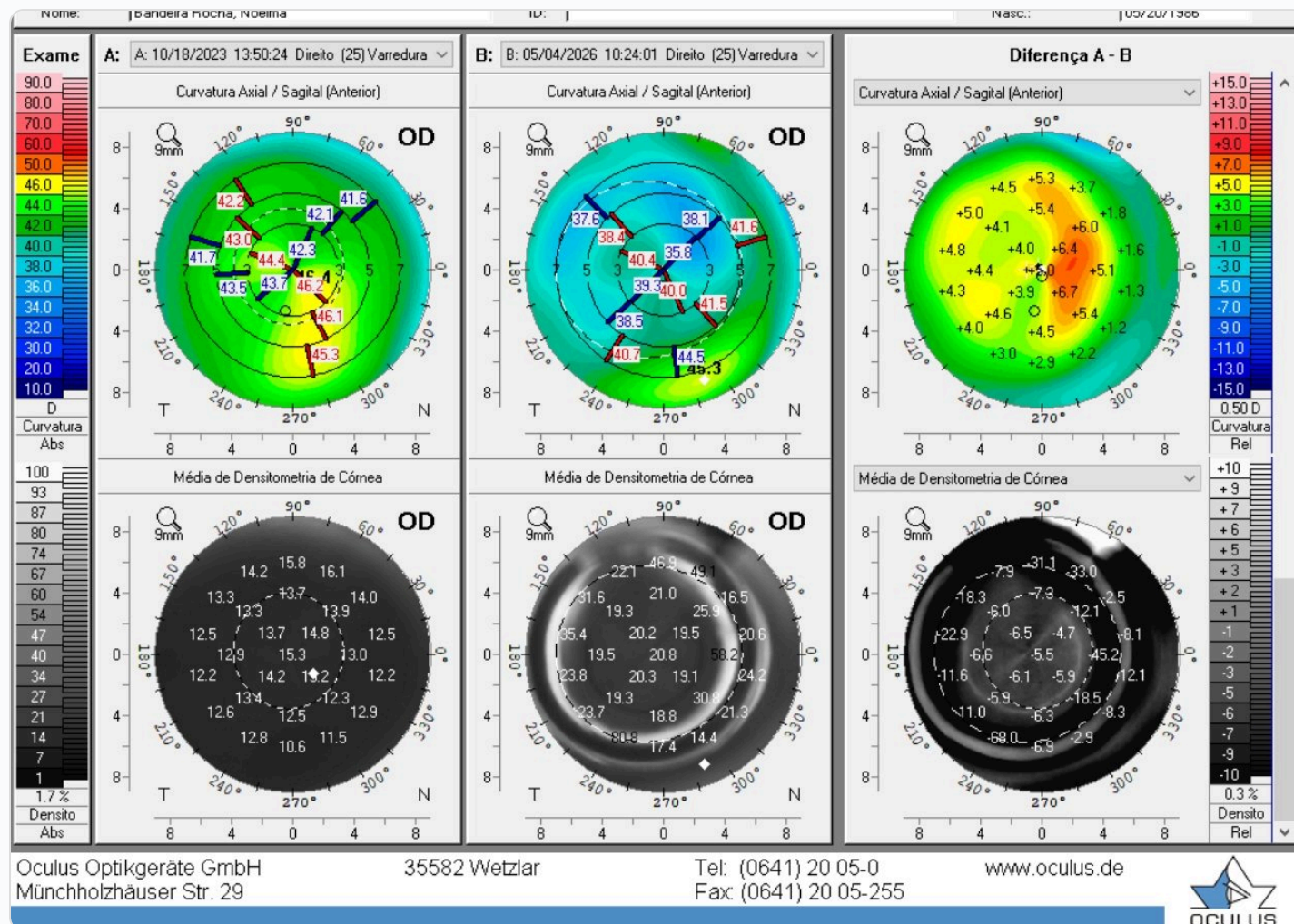
PARAMETER	Δ PRÉ→3M	Δ 3M→9M	Δ TOTAL (9M)
K1	1.47	2.85	4.32 D
K2	2.09	3.27	5.36 D
KM	1.92	3.01	4.94 D
Kmax	0.35	4.70	5.05 D
Q	0.42	0.37	0.79
Pachymetry	+9.58 μm	+33.19 μm	+42.77 μm

Performance by Model (3m PO)

MODEL	N	ΔK1	ΔKMAX	ΔASTIG
HM 300	23	2.92	6.92 D ★	—
HM 350	6	3.75	4.58 D	1.13
HM 210	3	2.04	4.81	1.40
HM 250	8	1.49	2.69	1.45
HM 400	5	0.60	1.91	1.60

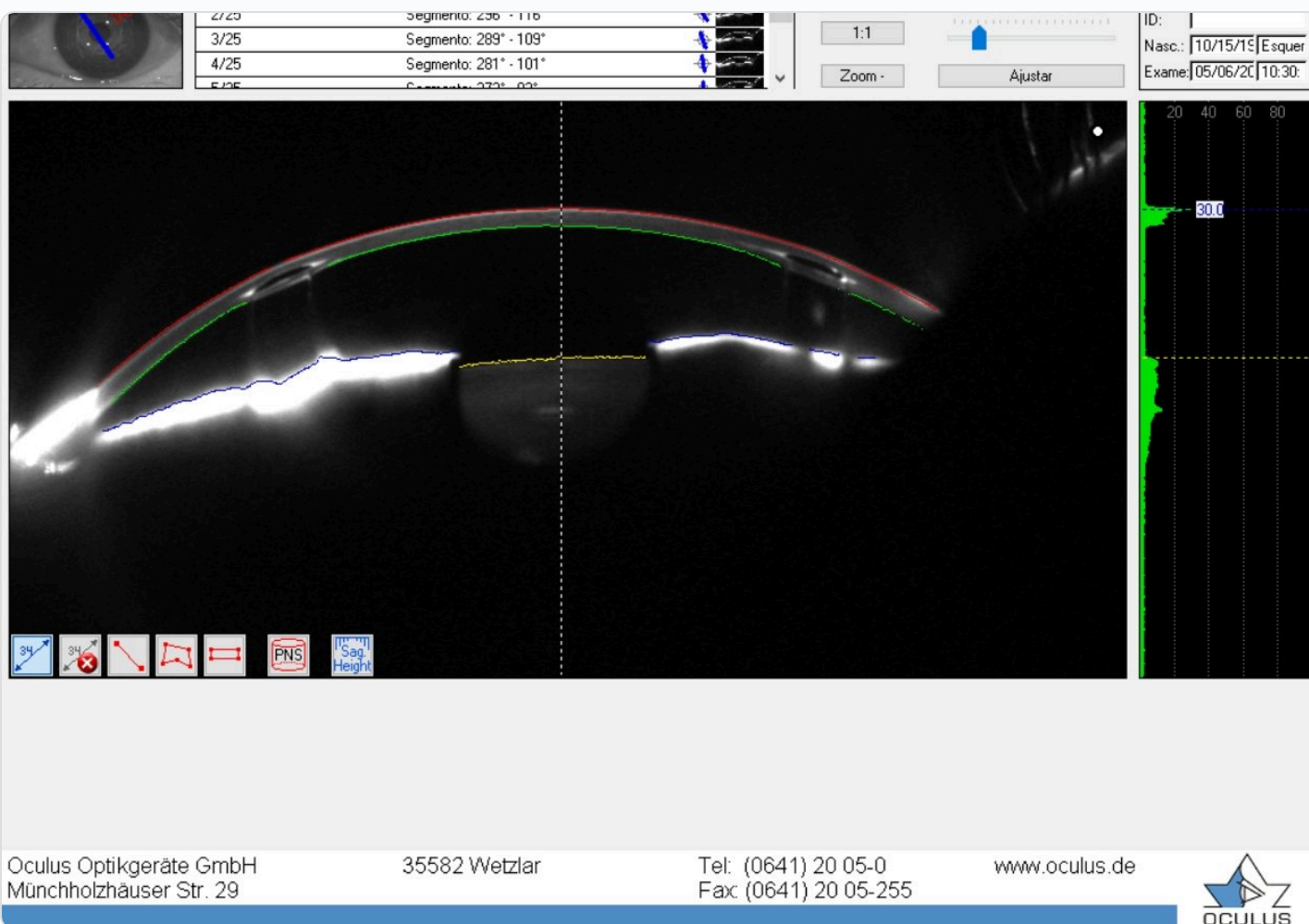
Clinical Cases

Case 1 — Keratoconus (OD) | Pentacam Compare



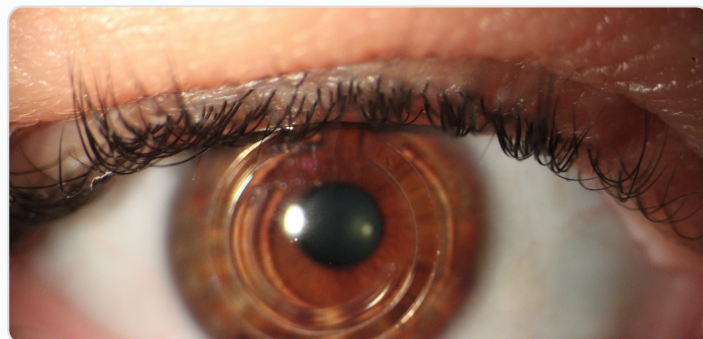
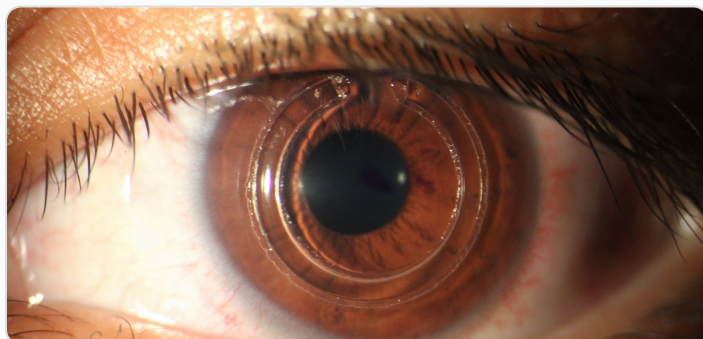
Anterior axial curvature: pre-op vs post-op. Differential map showing +3.9 to +6.7 D flattening in the cone zone.

Scheimpflug — HM Ring In Situ



Scheimpflug cross-section showing HM ring in the mid-stroma with preserved fusiform profile.

Slit-lamp — Implanted HM Ring



Corneal transparency preserved. Adequate centration. No deposits or neovascularization.

Competitive Advantages of Ferrara® HM



Exclusive Fusiform Profile

Exclusive fusiform design with optimized pressure distribution. 320° arc, 5.7 mm diameter.



Progressive Remodeling

93% of flattening after 3m. Sustained biomechanical effect up to 9+ months.



Concentric Combinations

Double-ring for advanced KC with enhanced centralized flattening.



Refractive Synergy

PRK/LASIK combination in selected cases for visual refinement.



Complete Range (200-450)

Coverage from mild to advanced KC + high myopia.

BR

100% Brazilian

World pioneer since 1986. Freedom manufacturing.

Conclusions

Progressive remodeling: ΔK_{max} 5.05 D at 9 months with intensification after 3m.

Tissue remodeling: +42.77 μm confirms active stromal remodeling.

HM 300 highlight: ΔK_{max} 6.92 D (n=23), above literature (2-4 D).

Versatility: Concentric combinations and refractive synergy expand indications.

References:

1. Ferrara P, et al. J Cataract Refract Surg. 2019.
2. Piñero DP, Alcón JL. Eye Vis. 2022.
3. Vega-Estrada A, Alió JL. J Refract Surg. 2016.